Ayoub GHRISS

Efficient ML, Sparsity & Quantization, GPU Systems

@ ayoub.ghriss@colorado.edu | +1 (720) 323-4395 | ayghri.me | in ayghri | Scholar | MA, FR

PUBLICATIONS

Preprint	S^3: Structured Sparsity Specification Ghriss, A. *Sparsity algebra unifying hardware-aligned patterns.*
Preprint	Kernelized Linear Attention: Breaking the Capacity Wall with Symmetric Cones Ghriss, A., Chakraborty, S. *Provable bounds on the capacity of linear attentions. Under review.*
AISTATS '26	Beyond Spectral Clustering: Probabilistic Cuts for Differentiable Graph Partitioning Ghriss, A. ***CU Boulder CS Dept. Outstanding Paper Award***
ICML-W '25	Crowdsourced Information Authentication: A Graph-based Model from the Science of Hadith Ghriss, A. *MusML Workshop, ICML 2025 — oral.*
AISTATS '25	Deep Clustering via Probabilistic Ratio-Cut Optimization Ghriss, A., Monteleoni, C. ***CU Boulder CS Dept. Outstanding Paper Award***
ASCE '24	Predicting Serious Injury and Fatality Exposure using Machine Learning in Construction Projects Oguz Erkal, E.D., Hallowell, M.R., Ghriss, A., Bhandari, S.
ICASSP '22	Sentiment-aware ASR Pre-training for Enhanced Speech Emotion Recognition Ghriss, A., Yang, B., Rozgic, V., Shriberg, E., Wang, C. ***SOTA on the valence (positivity) dimension at time of publication.***
NeuroImage '22	Insights from an Autism Imaging Biomarker Challenge: Promises and Threats to Biomarker Discovery Traut, N., et al. (incl. Ghriss, A.) *Best ML model among participants.*
RIKEN AIP'18	Reinforcement Learning with Options and State Representation Ghriss, A., Sugiyama, M., Lazaric, A. *MVA Master's thesis.*

SELECTED OPEN-SOURCE SOFTWARE

PGCuts: Differentiable graph cuts with GPU-accelerated hypergeometric kernels.	pgcuts.readthedocs.io
SparseKit: Structured sparsity specification (S^3) toolkit, with Structured-OBS for LLM pruning.	sparsekit.readthedocs.io
Qwantize: Optimal quantization for block-scaled low-precision formats (NVFP4, MXFP4).	qwantize.readthedocs.io
Tritonix: Triton GPU kernels with an optimized autotuning framework.	ayghri/tritonix
Sphedron: Refinable polyhedral meshes on the sphere for GNN-based geospatial ML.	sphedron.readthedocs.io

WORK EXPERIENCE

Teaching Assistant — Aug 2021 – Present
CU Boulder, Computer Science
- Graduate Machine Learning (CSCI 5266 / 5622): Fall '21, Fall '23, Spring '26.
- Undergraduate Machine Learning (CSCI 4266), '22–'23; guest instructor for the Reinforcement Learning module.
- Tutorials on Small Language Models (Fall '23).
- **Outstanding Teaching Assistant Award**, '22.

Graduate Research Assistant — Jan 2024 – Dec 2025
CIRES, CU Boulder
- GNN-based emulators for climate-model dynamics on refinable spherical meshes; released as the *Sphedron* library.

Visiting Researcher — Jun – Aug 2023
Inria, Paris
- Deep learning for optimization over discrete sets.

Applied Scientist Intern — May – Aug 2022
Amazon Alexa Paralinguistics, Cambridge
- Speaker identification and paralinguistic speech-emotion modeling on conversational audio.

Applied Scientist Intern — May – Aug 2021
Amazon Alexa Edge-ML, Cambridge
- Sentiment-aware ASR pre-training for on-device speech-emotion recognition (ICASSP '22).

Software Engineer, PhD Intern — Jun – Aug 2020
Google, Boulder
- Deep learning for air-quality prediction from satellite imagery (Google Earth Engine).

Research Intern — Apr – Sep 2018
RIKEN Advanced Intelligence Project, Tokyo
- Hierarchical deep reinforcement learning with options and learned state representations.

SERVICE & TALKS

- **Reviewer:** ICLR 2025 • ICML 2025, 2026 • NeurIPS 2025
- **Tutorial Chair:** "Associative Memories", ICML 2025
- **Oral Presentation:** *Crowdsourced Information Authentication: A Graph-based Model from the Science of Hadith*, MusML Workshop, ICML 2025
- **Workshop Instructor:** *GPU Programming Workshop* — CS Graduate Professional Development Club, CU Boulder, Spring '26. Hands-on intro to parallel execution, thread/warp behavior, fused kernels, and Triton. github.com/ayghri/gpuwork
- **Guest Lectures:** *Introduction to Machine Learning* — ASEN 5307 Engineering Data Analysis Methods, CU Boulder (2 lectures in Fall '24, 2 in Fall '25)

AWARDS & HONORS

- **CU Boulder CS Dept. Outstanding Paper Award** — for *Beyond Spectral Clustering: Probabilistic Cuts for Differentiable Graph Partitioning* (AISTATS '26).
- **CU Boulder CS Dept. Outstanding Paper Award** — for *Deep Clustering via Probabilistic Ratio-Cut Optimization* (AISTATS '25).
- **Best Work-in-Progress Award**, CU Boulder CS Dept. Expo, Fall '25 — for work on structured sparse training (S^3).
- **Outstanding Teaching Assistant Award**, CU Boulder CS Dept., '22.

EDUCATION

PhD, Machine Learning 2019 – 2026
University of Colorado, Boulder
- Advisor: Claire Monteleoni.
- Thesis: Structured Sparse Training and Differentiable Graph Cuts for Efficient ML.

Research Master — Mathematics, Computer Vision, and Machine Learning (MVA) Oct 2017 – Sep 2018
École Normale Supérieure Paris-Saclay
- Reinforcement Learning, Probabilistic Graphical Models, Statistical Learning
- Computer Vision, Deep Learning, Speech Recognition, Audio Signal Processing

Diplôme d'Ingénieur — Polytechnicien cycle Sep 2014 – Sep 2017
École Polytechnique
- Polytechnician Engineer Cycle with Major in Applied and Computational Mathematics.
- Stochastic Calculus, Monte Carlo methods, Advanced Machine Learning, Random Models of survival.

B.Sc. Economics and Applied Mathematics Sep 2013 – Jun 2014
ENSAE ParisTech
- Data Analysis, Econometrics, Microeconomics, Macroeconomics

LANGUAGES

Arabic	Native
French	Native
English	Fluent
Japanese	Intermediate
Chinese	Elementary

TECHNICAL SKILLS

GPU & Systems CUDA, Triton, C/C++, Linux, Git

ML / DL Python, PyTorch, Scikit-learn, SciPy

Distributed AWS, PySpark, Horovod, MPI